

Main: HomePage



INDIVIDUAL PROJECT

TSA2151 - SYSTEM ADMINISTRATION AND MAINTENANCE TRIMESTER 1, 2023/2024 (TERM2310)

Lecture Session: SA1

Submission Date: 19 Jan 2024

NO	STUDENT ID.	STUDENT NAME	MAJOR
1.	1221302732	HANI HAMIZAH BINTI HAMZAN	ST

From PmWiki

Main: TableOfContents

No.	Content	Page
0.	Table Of Contents	i.
1.	Introduction To Linux	1
2.	Managing Users	2
3.	Installing Internet Server	4
4.	Installing And Configuring CMS	7
5.	Installing And Configuring PmWiki	10

Retrieved from <http://192.168.56.105/pmwiki/index.php?n=Main.TableOfContents>
Page last modified on January 21, 2024, at 02:03 pm

From PmWiki

Main: MainContent

Introduction To Linux

Linux is a flexible operating system that is compatible with almost all types of devices. While other operating systems such as Windows and MacOS are either closed source or partially closed source, Linux Distribution is mostly open source. The operating system is a system that manages the communication between the end user and the hardware.

Linux is mainly composed of several different pieces:

1. **Bootloader** – The software that manages the boot process of your computer. For most users, this will simply be a splash screen that pops up and eventually goes away to boot into the operating system.
2. **Kernel** – This is the one piece of the whole that is called ‘Linux’. The kernel is the core of the system and manages the CPU, memory, and peripheral devices. The kernel is the lowest level of the OS.
3. **Init system** – This is a sub-system that bootstraps the user space and is charged with controlling daemons. One of the most widely used init systems is systemd, which also happens to be one of the most controversial. It is the init system that manages the boot process, once the initial booting is handed over from the bootloader (i.e., GRUB or GRand Unified Bootloader).
4. **Daemons** – These are background services (printing, sound, scheduling, etc.) that either start up during boot or after you log into the desktop.
5. **Graphical server** – This is the sub-system that displays the graphics on your monitor. It is commonly referred to as the X server or just X.
6. **Desktop environment** – This is the piece that the users interact with. There are many desktop environments to choose from (GNOME, Cinnamon, Mate, Pantheon, Enlightenment, KDE, Xfce, etc.). Each desktop environment includes built-in applications (such as file managers, configuration tools, web browsers, and games).
7. **Applications** – Desktop environments do not offer the full array of apps. Just like Windows and macOS, Linux offers thousands upon thousands of high-quality software titles that can be easily found and installed. Most modern Linux distributions (more on this below) include App Store-like tools that centralize and simplify application installation. For example, Ubuntu Linux has the Ubuntu Software Center (a rebrand of GNOME Software) which allows you to quickly search among the thousands of apps and install them from one centralized location.

There are a few popular Linux distributions that are built for a variety level of users from beginner to hard-core users.

- LINUX MINT
- MANJARO
- DEBIAN
- UBUNTU
- ANTERGOS
- SOLUS
- FEDORA
- ELEMENTARY OS
- OPENSUSE

Introduction and Installing Pop!_OS

PopOS is a Ubuntu-based Linux operating System that is built specially for STEM and creative professionals that use their computer as a tool to work and create and yes, this is one of hundreds of open-sourced Linux distributions. There are many suitable built-in libraries and tools in Pop_OS! As this operating system is Ubuntu-based, Pop!_OS Uses APT and Flatpak package management which makes it easy to install, remove and update all software on the OS. Pop!_OS is easy to handle and only available in the desktop version, featuring a customized GNOME desktop environment named COSMIC. It is developed by American Linux computer manufacturer, System76.

Pop!_OS provides full out-of-the-box support for both AMD and Nvidia GPUs. A 2020 review in Foss BYTES described it as an easy distribution to set up for gaming, mainly due to its built-in GPU support. Pop!_OS provides default disk encryption, streamlined window and workspace management, keyboard shortcuts for navigation as well as built-in power management profiles. The latest releases also have packages that allow for easy setup for TensorFlow and CUDA.

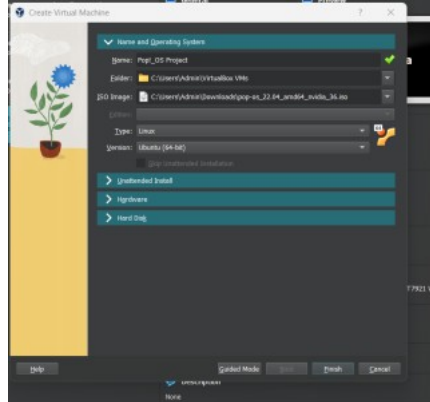
Installation can be done at link text: [Pop!_OS Install](#)



It is worth noting that installation must be done according to your GPU driver for the best experience.



After finishing the ISO installation, we can open Oracle VM VirtualBox to New > Name and add ISO image



Wait for a moment until the GUI of Pop!_OS Appears Choose the Language and time zone, and opt for a clean install to ensure all tools are available.



There will be a partitioning disk and you can check your desktop environment by writing this command on the terminal

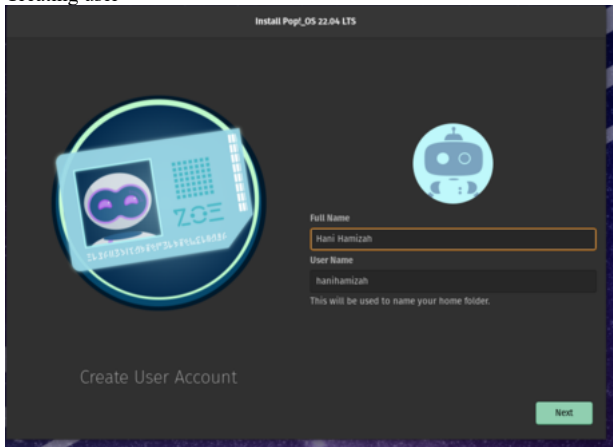
```
haniahamizah@pop-os: ~$ echo $XDG_CURRENT_DESKTOP
pop:GNOME
```

Managing Users

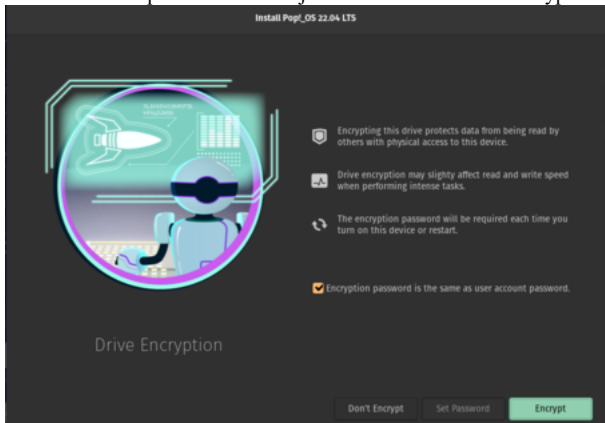
A user is an entity, in a Linux operating system, that can manipulate files and perform several other operations. Each user is assigned an ID that is unique for each user in the operating system. In this post, we will learn about users and commands which are used to get information about the users.

After installation of the operating system, the ID 0 is assigned to the root user, and the IDs 1 to 999 (both inclusive) are assigned to the system users hence the IDs for local users begin from 1000 onwards. Usually, the sysadmin needs to create a user as an admin on the terminal. However, Pop!_OS is installed with GUI so we can create a user after the clean install is finished.

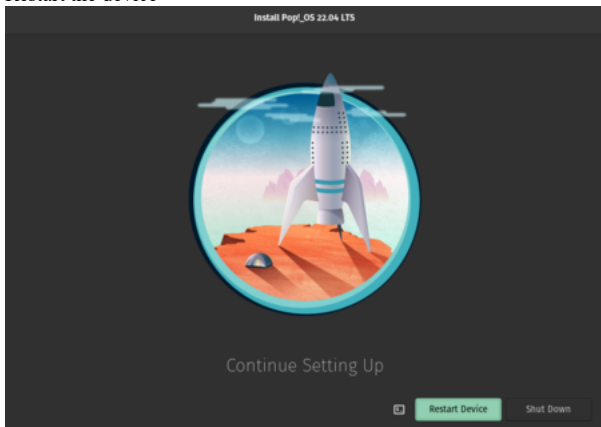
1. Creating user



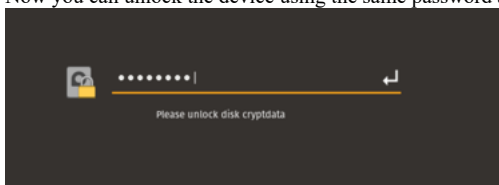
2. The main user's password created just now will be used to encrypt the disk



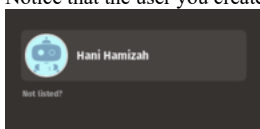
3. Restart the device



4. Now you can unlock the device using the same password as the main user just now.



5. Notice that the user you created just now is available in the user list.



6. Open the terminal and write create the first user, John Doe with password sysadmin#1

```
hanizah@pop-os:~$ sudo adduser jdoe
[sudo] password for hanizah:
Adding user `jdoe' ...
Adding new group `jdoe' (1001) ...
Adding new user `jdoe' (1001) with group `jdoe' ...
Creating home directory `/home/jdoe' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for jdoe
Enter the new value, or press ENTER for the default
  Full Name []: John Doe
    Room Number []:
    Work Phone []:
    Home Phone []:
      Other []:
Is the information correct? [Y/n] Y
```

7. You can opt to add user John Doe as a super user into the sudo group.

```
hanizah@pop-os:~$ sudo adduser jdoe sudo
[sudo] password for hanizah:
Adding user `jdoe' to group `sudo' ...
Adding user jdoe to group sudo
Done.
```

8. Now add the second additional user, Will Smith with password sysadmin#2 using the same command just now.

```
hanizah@pop-os:~$ sudo adduser wsmith
Adding user `wsmith' ...
Adding new group `wsmith' (1002) ...
Adding new user `wsmith' (1002) with group `wsmith' ...
Creating home directory `/home/wsmith' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for wsmith
Enter the new value, or press ENTER for the default
  Full Name []: Will Smith
    Room Number []:
    Work Phone []:
    Home Phone []:
      Other []:
Is the information correct? [Y/n]
```

9. Now you can use the command `ls /home` to check the list of users created.

```
hanizah@pop-os:~$ ls /home
hanizah  jdoe  wsmith
```

10. You can login into John Doe home directory using this command `su -l username`. Notice that the username@pop-os has changed.

```
hanizah@pop-os:~$ su -l jdoe
Password:
jdoe@pop-os:~$
```

11. You can also login into Will Smith home directory using this command `su -l username`. Notice that the username@pop-os has changed.

```
jdoe@pop-os:~$ su -l wsmith
Password:
wsmith@pop-os:~$
```

Installing Internet Server

FTP

FTP servers are the software solutions used for transferring files across the internet. They are primarily used for two essential functions, “Put” and “Get.” It allows uploading (Put) files to the server from the client device and downloading (Get) files from the server on the client device. FTP server helps to accommodate the following functions.

To install ftp, follow the following command

1. We are using apt-get install proftpd-basic as Pop! OS is a Ubuntu-based sudo apt-get install proftpd-basic

```
hanizah@pop-os:~$ sudo apt-get install proftpd-basic
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libhiredis0.14 libmemcached11 libmemcachedutil2 proftpd-core proftpd-doc
  proftpd-mod-crypto proftpd-mod-wrap
Suggested packages:
  openssl-inetd | inet-superserver proftpd-mod-ldap proftpd-mod-mysql
  proftpd-mod-odbc proftpd-mod-openssl proftpd-mod-sqlite proftpd-mod-geoip
  proftpd-mod-sftp
The following NEW packages will be installed:
  libhiredis0.14 libmemcached11 libmemcachedutil2 proftpd-basic proftpd-core
  proftpd-doc proftpd-mod-crypto proftpd-mod-wrap
0 upgraded, 8 newly installed, 0 to remove and 193 not upgraded.
Need to get 3,485 kB of archives.
After this operation, 11.8 MB of additional disk space will be used.
```

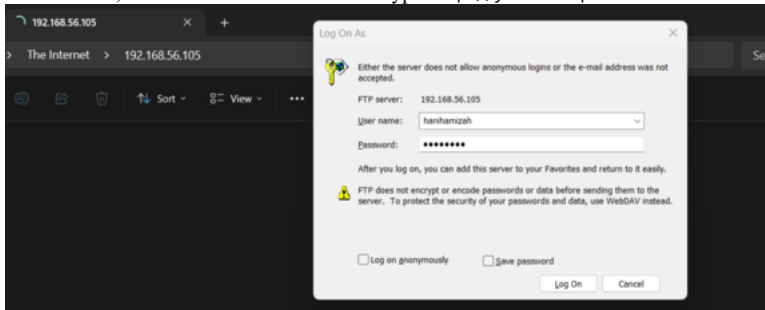
2. For this Ubuntu-based operating system, as we install ftpd-basic it is auto-activated but not connected we can check whether it is connected or not by using `ftp` command

```
hanizah@pop-os:~$ ftp
ftp>
ftp> status
Not connected.
No proxy connection.
Gate ftp: off, server (none), port ftpgate.
Passive mode: on; fallback to active mode: on.
Mode: ; Type: ; Form: ; Structure: .
Verbose: on; Bell: off; Prompting: on; Globbing: on.
Store unique: off; Receive unique: off.
Preserve modification times: on.
Case: off; CR stripping: on.
Wtrans: off.
Mmap: off.
Hash mark printing: off; Mark count: 1024; Progress bar: on.
Get transfer rate throttle: off; maximum: 0; increment 1024.
Put transfer rate throttle: off; maximum: 0; increment 1024.
Socket buffer sizes: send 0, receive 0.
Use of PORT cmds: on.
Use of EPSV/EPRT cmds for IPv4: on.
Use of EPSV/EPRT cmds for IPv6: on.
Command line editing: on.
Version: tnftp 20210827
ftp>
```

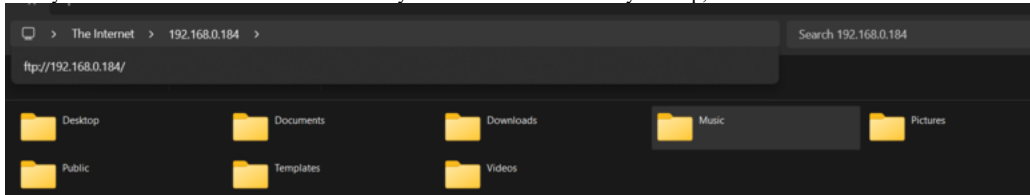
3. To connect ftp server with localhost, use the command `ftp localhost` and key in your directory username and password

```
hanihamizah@pop-os:~$ ftp pop-os
Trying 192.168.0.184:21 ...
Connected to pop-os.
220 ProFTPD Server (Debian) [::ffff:192.168.0.184]
Name (pop-os:hanihamizah): hanihamizah
331 Password required for hanihamizah
Password:
230 User hanihamizah logged in
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

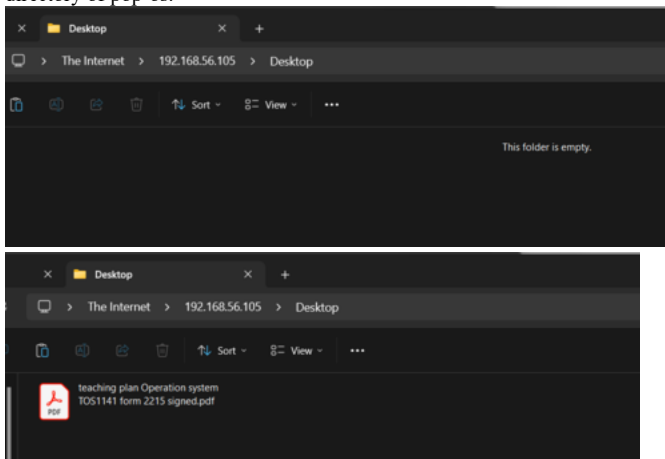
4. To connect it, we can use Windows File and type `ftp://yourhostipaddress`



5. Now you can see the file available in the host you are connected to. To try this ftp, we can transfer files from Windows to the connected host just now.



6. Notice how Desktop file is initially empty, and then we copied a file in the desktop from Windows using FTP. Now we can see the file appears in the desktop directory of pop-os.



SSH

SSH, also known as Secure Shell or Secure Socket Shell, is a network protocol that gives users, particularly system administrators, a secure way to access a computer over an unsecured network.

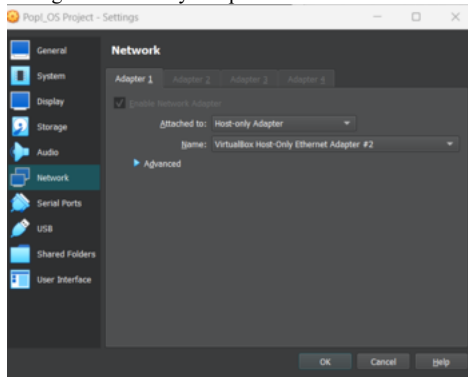
SSH also refers to the suite of utilities that implement the SSH protocol. Secure Shell provides strong password authentication and public key authentication, as well as encrypted data communications between two computers connecting over an open network, such as the Internet.

In addition to providing strong encryption, SSH is widely used by network administrators to manage systems and applications remotely, enabling them to log in to another computer over a network, execute commands, and move files from one computer to another. Here we will learn how to install and configure SSH on Pop!_OS

1. To install it, we use `sudo apt-get install openssh-server`

```
hanihamizah@pop-os:~$ sudo apt-get install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  openssh-client
1 upgraded, 4 newly installed, 0 to remove and 192 not upgraded.
Need to get 1,657 kB of archives.
After this operation, 6,859 kB of additional disk space will be used.
Do you want to continue? [y/n]
Get:1 http://apt.pop-os.org/ubuntu jammy-security/main amd64 openssh-client
```

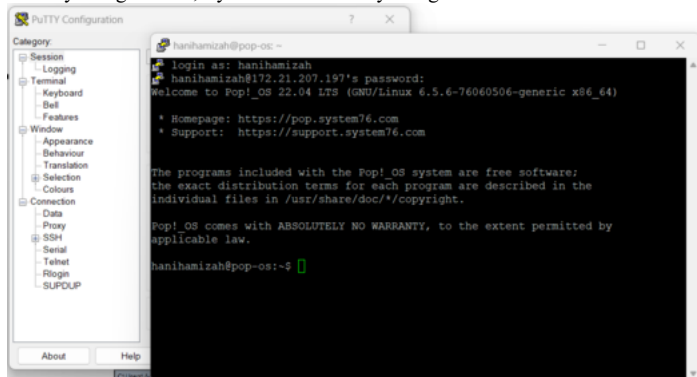
2. Change to Host-Only Adapter on virtual machine network



3. Check one more time the IP address of host `hostname -I`

```
hanimamizah@pop-os:~$ hostname -I
192.168.0.184
hanimamizah@pop-os:~$
```

4. Now by using PuTTY, try the SSH server by using IP address of our virtual machine and username



LAMP (Linux, Apache, MySQL, PHP)

A LAMP server is a configuration setup commonly used for web development and hosting dynamic web applications. Noted that LAMP-server might not be available to some Linux Distribution other than Ubuntu. The term "LAMP" is an acronym that stands for the four main components of this server setup:

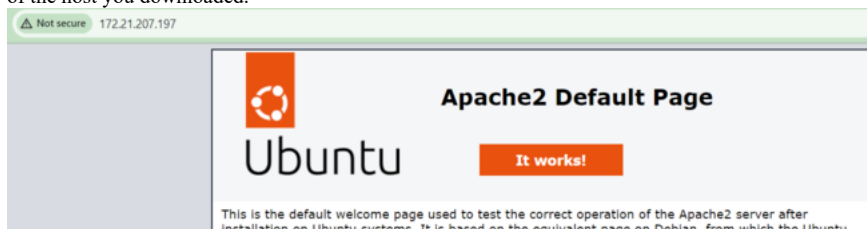
1. Linux: The operating system on which the server runs. Linux is a popular choice due to its stability, security features, and open-source nature.
2. Apache: The web server software that handles HTTP requests and serves web pages. Apache is widely used and known for its reliability and flexibility. It can interpret server-side scripting languages, making it suitable for dynamic web applications.
3. MySQL: The relational database management system (RDBMS) used for storing and managing data. MySQL is often used to store data for web applications and is known for its speed and reliability.
4. PHP, Python, or Perl: The programming language used to develop server-side scripts that generate dynamic web content. PHP is the most commonly used language in the LAMP stack, but Python and Perl are also compatible.

Step to install are as follows:

1. Download from terminal using command `sudo apt-get install lamp-server^`

```
hanimamizah@pop-os:~$ sudo apt-get install lamp-server^
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'libevent-pthreads-2.1-7' for task 'lamp-server'
Note, selecting 'libperl5.34' for task 'lamp-server'
Note, selecting 'libcurl4' for task 'lamp-server'
Note, selecting 'libtime-date-perl' for task 'lamp-server'
Note, selecting 'libhtml-tagset-perl' for task 'lamp-server'
Note, selecting 'libgdbm-compat4' for task 'lamp-server'
Note, selecting 'libssh-4' for task 'lamp-server'
Note, selecting 'ssl-cert' for task 'lamp-server'
Note, selecting 'libfcgi-perl' for task 'lamp-server'
Note, selecting 'perl' for task 'lamp-server'
Note, selecting 'libldap-common' for task 'lamp-server'
Note, selecting 'libaio1' for task 'lamp-server'
Note, selecting 'libaprutil1-dbd-sqlite3' for task 'lamp-server'
Note, selecting 'mysql-server-8.0' for task 'lamp-server'
Note, selecting 'libapr1' for task 'lamp-server'
```

2. Now we can check if Apache is successfully installed or not by going to the web browser and typing `localhost` or replacing local host with the current IP address of the host you downloaded.



3. Next, we are going to enable info.php to check if our php is successfully installed on our server.

1. To do so, we must go to /var/www/html directory and create info.php file if it is not there yet using `sudo nano info.php`

```
hanishamizah@pop-os:~$ cd /var/www/html
hanishamizah@pop-os:/var/www/html$ ls
index.html
hanishamizah@pop-os:/var/www/html$ nano info.php
hanishamizah@pop-os:/var/www/html$ ls
index.html
hanishamizah@pop-os:/var/www/html$ sudo nano info.php
hanishamizah@pop-os:/var/www/html$ ls
index.html  info.php
```

2. Edit info.php file using `nano` with `<?php phpinfo(); ?>` > CTRL + X > CTRL+Y > CTRL M

```
hanishamizah@pop-os: /var/www/html
GNU nano 6.2 info.php
<?php phpinfo(); ?>
```

3. Reboot the machine to initialize the server with the new configuration.

```
hanishamizah@pop-os:~$ reboot
```

4. Now we can check on web browser if `ipaddress/info.php` page appears



Installing And Configuring CMS

A Content Management System (CMS) is a software application or set of related programs that enables users to create, manage, and publish digital content on the web. A CMS provides an intuitive and user-friendly interface for individuals, even those without extensive technical knowledge, to collaboratively create and update content.

Key features and functionalities of a CMS typically include:

- **Content Creation and Editing:** Users can create, edit, and format digital content (text, images, videos, etc.) using a built-in editor that often resembles word processing software.
- **Content Storage and Organization:** The CMS stores content in a structured manner, often using a database. Content is organized into categories, tags, or folders to facilitate easy retrieval and management.
- **User Management and Permissions:** CMSs allow multiple users to collaborate on content creation and management. User roles and permissions control who can view, edit, or publish content, ensuring security and workflow control.
- **Version Control:** Many CMSs provide version control features, allowing users to track changes, revert to previous versions, and collaborate on content without the risk of overwriting each other's work.

Examples of popular CMSs include:

- WordPress
- Joomla
- Drupal
- Magento (for e-commerce)
- Wix
- Squarespace

Today we are going to install WordPress. The steps are shown as follows:

1. Go to /var/www/html directory by using the command `cd /var/www/html` on the terminal.

2. To install it using terminal, type the command `sudo wget https://wordpress.org/latest.zip` and unzip it using command `sudo unzip filename`

```
hanishamizah@pop-os:/var/www/html$ sudo wget https://wordpress.org/latest.zip
[sudo] password for hanishamizah:
--2024-01-11 21:24:37-- https://wordpress.org/latest.zip
Resolving wordpress.org (wordpress.org)... 198.143.164.252
Connecting to wordpress.org (wordpress.org)|198.143.164.252|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 25954973 (25M) [application/zip]
Saving to: 'latest.zip'

latest.zip  100%[=====] 24.75M  2.29MB/s  in 12s
2024-01-11 21:24:11 (2.04 MB/s) - 'latest.zip' saved [25954973/25954973]

hanishamizah@pop-os:/var/www/html$ ls
index.html  info.php  latest.zip  wp-ai
hanishamizah@pop-os:/var/www/html$ sudo unzip latest.zip
Archive: latest.zip
  creating: wordpress/
  inflating: wordpress/xmlrpc.php
  inflating: wordpress/wp-blog-header.php
  inflating: wordpress/readme.html
```

3. After that, we need to create a database for WordPress using MySQL downloaded from lamp-server just now.

```
hanishamizah@pop-os:/var/www/html$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.35-0ubuntu0.22.04.1 (Ubuntu)

Copyright (c) 2000, 2023, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE wordpress;
Query OK, 1 row affected (0.03 sec)
```

4. Create a user for the WordPress database so that the user can access to WordPress database

```
mysql> CREATE USER 'user'@'localhost' IDENTIFIED BY 'user123';
Query OK, 0 rows affected (0.05 sec)

mysql> GRANT ALL ON wordpress.* TO 'user'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> exit
```

5. Next we are configuring WordPress to make it available on our server. Firstly, we go to the WordPress directory and find the configuration file. If there is no `wp-config.php` then we must create the file. This file will be treated as the default configuration of WordPress on our Server.

```
hanishamizah@pop-os: /var/www/html/wordpress$ ls
index.php      wp-blog-header.php  wp-includes      wp-settings.php
license.txt    wp-comments-post.php wp-links-opml.php wp-signup.php
readme.html   wp-config-sample.php wp-load.php       wp-trackback.php
wp-activate.php wp-content           wp-login.php      xmlrpc.php
wp-admin       wp-cron.php          wp-mail.php
hanishamizah@pop-os: /var/www/html/wordpress$ sudo touch wp-config.php
hanishamizah@pop-os: /var/www/html/wordpress$ ls
index.php      wp-blog-header.php  wp-cron.php       wp-mail.php
license.txt    wp-comments-post.php wp-includes        wp-settings.php
readme.html   wp-config.php       wp-links-opml.php wp-signup.php
wp-activate.php wp-config-sample.php wp-load.php        wp-trackback.php
wp-admin       wp-content          wp-login.php      xmlrpc.php
```

6. Copy the contents of the sample configuration file into `wp-config.php` created just now.

```
hanishamizah@pop-os: /var/www/html/wordpress$ sudo cp wp-config-sample.php wp-config.php
hanishamizah@pop-os: /var/www/html/wordpress$ nano wp-config.php
```

7. Edit `wp-config.php` by uncommenting the configuration file and enabling the Database name, user, host, and password according to the database created just now.

```
hanishamizah@pop-os: /var/www/html/wordpress$ nano wp-config.php
GNU nano 6.2 wp-config.php
/*
 * WordPress database configuration
 */

// ** Database settings - You can get this info from your web host ** //
/* The name of the database for WordPress */
define( 'DB_NAME', 'wordpress' );

/* Database username */
define( 'DB_USER', 'user' );

/* Database password */
define( 'DB_PASSWORD', 'user123' );

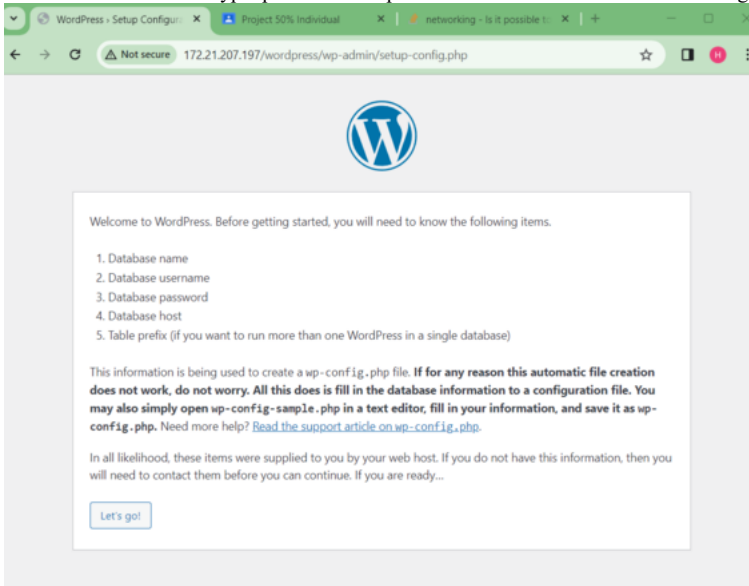
/* Database hostname */
define( 'DB_HOST', 'localhost' );

/* Database charset to use in creating database tables. */
define( 'DB_CHARSET', 'utf8' );

/* The database collate type. Don't change this if in doubt. */
define( 'DB_COLLATE', '' );

/* Help
 * Exit
 * Write Out
 * Read File
 * Where Is
 * Replace
 * Cut
 * Paste
 * Execute
 * Justify
 * Location
 * Go To Line
 */
```

8. Go to Web Browser and type `ipaddress/wordpress` to ensure WordPress is now working and connected to our server.



9. Fill in the information needed for database connected to wordpress. This information **must** be the same as the created database we did in MySQL and we set on the configuration file.

Below you should enter your database connection details. If you are not sure about these, contact your host.

Database Name
wordpress
The name of the database you want to use with WordPress.

Username
user
Your database username.

Password
user123
Your database password.

Database Host
localhost
You should be able to get this info from your web host, if localhost does not work.

Table Prefix
wp_
If you want to run multiple WordPress installations in a single database, change this.

[Submit](#)

10. Fill in the information needed. This information will be used to log in to Wordpress.

Welcome

Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title
System Administration and Maintenance

Username
admin
Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password
IY8B8oHbLWJ15(3J8H
Strong

Important: You will need this password to log in. Please store it in a secure location.

Your Email
1221302732@student.mmu.edu.my
Double-check your email address before continuing.

Search engine visibility
☐ Discourage search engines from indexing this site
It is up to search engines to honor this request.

[Install WordPress](#)

11. If there is no problem or error, you are supposed to get to this page saying **Success!**

Success!

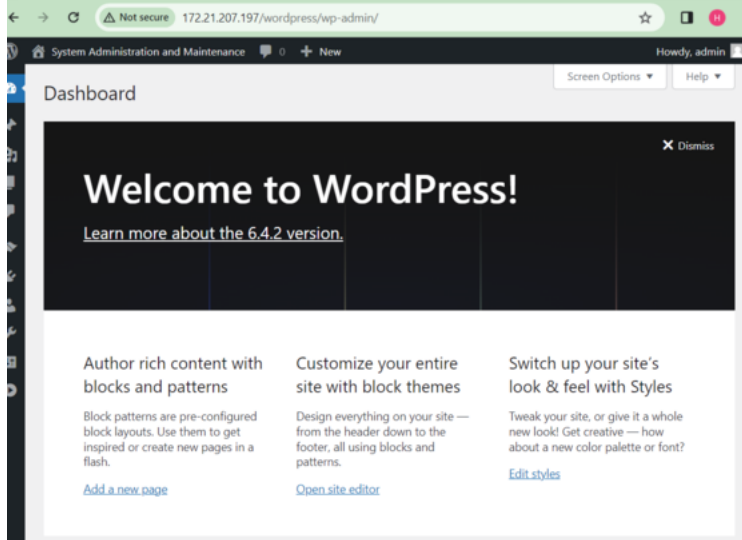
WordPress has been installed. Thank you, and enjoy!

Username
admin

Password
Your chosen password.

[Log In](#)

12. Now we will be in the dashboard of WordPress when going to address `ipaddress/wordpress` on a web browser.



Installing And Configuring PmWiki

Wiki basically is a website or database developed collaboratively by a community of users, allowing any user to add and edit content. The most popular wiki is Wikipedia, it uses Mediawiki system. Many wiki system available for free and you can set up it easily in your web server including mediawiki, dokuwiki, pmwiki and others. Setup and configure pmwiki system in your Apache web server.

Follow the following steps to install and configure PmWiki

1. Go to directory `/var/www/html` using command `cd /var/www/html/`

2. Download PmWiki zip file on terminal using command `sudo wget http://www.pmwiki.org/pub/pmwiki/pmwiki-latest.zip`

```
hanishamirah@pop-os: /var/www/html$ sudo wget http://www.pmwiki.org/pub/pmwiki/pmwiki-latest.zip
URL transformed to HTTPS due to an HSTS policy
--2024-01-11 18:10:00-- https://www.pmwiki.org/pub/pmwiki/pmwiki-latest.zip
Resolving www.pmwiki.org (www.pmwiki.org)... 23.254.203.248
Connecting to www.pmwiki.org (www.pmwiki.org)[23.254.203.248]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 663932 (648K) [application/zip]
Saving to: 'pmwiki-latest.zip'

pmwiki-latest.zip  100%[=====] 648.37K  279KB/s  in 2.3s

2024-01-11 18:10:04 (279 KB/s) - 'pmwiki-latest.zip' saved [663932/663932]
```

3. If there is no error and successfully downloads, we can unzip the file. As we unzip the file, there will be a directory name `pmwiki-2.3.29`

```
hanishamirah@pop-os: /var/www/html/pmwiki$ sudo unzip pmwiki-latest
Archive: pmwiki-latest.zip
creating: pmwiki-2.3.29/
creating: pmwiki-2.3.29/cookbook/
inflating: pmwiki-2.3.29/cookbook/.htaccess
inflating: pmwiki-2.3.29/pmwiki.php
creating: pmwiki-2.3.29/pub/
creating: pmwiki-2.3.29/pub/skins/
creating: pmwiki-2.3.29/pub/skins/pmwiki-responsive/
inflating: pmwiki-2.3.29/pub/skins/pmwiki-responsive/README
inflating: pmwiki-2.3.29/pub/skins/pmwiki-responsive/skin.js
inflating: pmwiki-2.3.29/pub/skins/pmwiki-responsive/skin.templ
inflating: pmwiki-2.3.29/pub/skins/pmwiki-responsive/skin.php
inflating: pmwiki-2.3.29/pub/skins/pmwiki-responsive/xmenu.svg
inflating: pmwiki-2.3.29/pub/skins/pmwiki-responsive/skin.css
```

4. Create new directory in `/var/www/html` named `pmwiki` using command `sudo mkdir pmwiki` and move `pmwiki-2.3.29` into `pmwiki` directory. Now we can go into `pmwiki` directory. Create one directory named `wiki.d` in `pmwiki` directory. This is directory is important because all data and metadata or the Wiki System will be stored here.

```
hanishamirah@pop-os: /var/www/html$ sudo mv pmwiki-2.3.29 pmwiki
hanishamirah@pop-os: /var/www/html$ cd pmwiki
hanishamirah@pop-os: /var/www/html/pmwiki$ sudo mkdir wiki.d
hanishamirah@pop-os: /var/www/html/pmwiki$ ls
cookbook docs local pmwiki.php pub README.txt scripts wiki.d wikilib.d
hanishamirah@pop-os: /var/www/html/pmwiki$
```

5. Next, change the system owner of PmWiki

```
hanishamirah@pop-os: /var/www/html/pmwiki$ sudo chown -R www-data:www-data /var/www/html/pmwiki
hanishamirah@pop-os: /var/www/html/pmwiki$
```

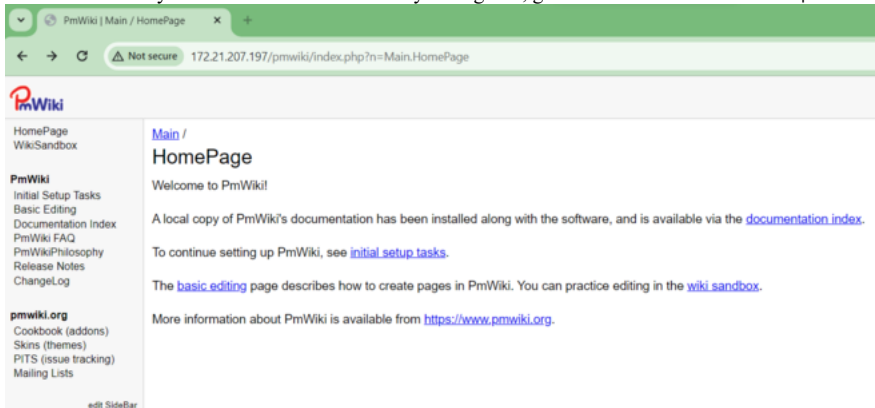
6. Go to `index.php` in PmWiki directory. If there is none, create one in PmWiki directory and copy this code. `<?php include ('pmwiki.php');` to embedded our PmWiki page with php code.

```
hanishamirah@pop-os: /var/www/html/pmwiki$ nano index.php
GNU nano 6.2 index.php
<?php include ('pmwiki.php');
```

7. Now we go copy sample configuration file in `docs` directory to our main configuration file in `/var/www/html/pmwiki/local/config.php` if there is no local file in `pmwiki`, create one directory named `local` `sudo mkdir local` and create configuration file using command `touch config.php`

```
hanishamirah@pop-os: /var/www/html/pmwiki$ sudo cp docs/sample-config.php local/config.php
```

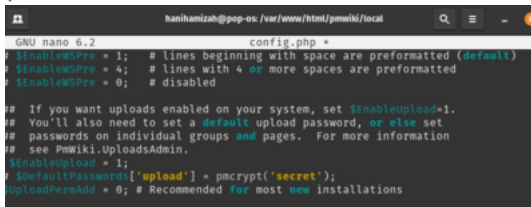
8. Now we can verify if our PmWiki is successfully configured, go to web browser and write `ipaddress/pmwiki` . You should get this front page as default page.



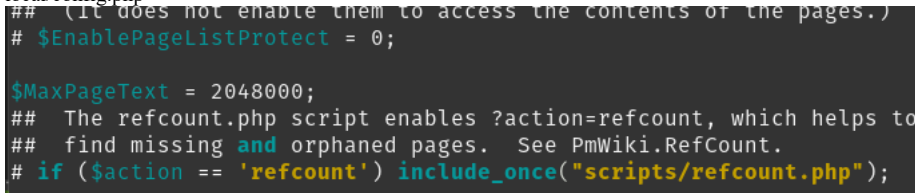
Extra Configuration For PmWiki

Configuration for Upload and Increase Maximum Text on One Page

1. Uploading on PmWiki is not enable by default so we must configure to enable the option in `local/config.php` file. Uncomment `$EnableUpload` and `$DefaultPassword`.



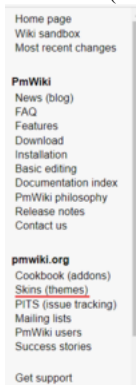
2. There will be some restriction for amount of text per page. To overcome this problem, we can add configure line `$MaxPageText = numberoftextperpage` in `local/config.php`



Configure PmWiki Theme.

PmWiki theme can be varied and available in PmWiki Skins. We can change it by doing some configuration in skins file. We can personalize the PmWiki appearance either using the skins downloaded or we make it ourselves using css and html.

1. Click Skins (themes) on the sidebar of PmWiki.



2. We can choose skins from the page and visit the link to see how it looks like.

Skins ↑

Name	Summary
2016 skin	Responsive skin very close
ABitModern	PmWiki port of the oswd.or
Academian	A tan skin with a gradient n
Adapt Skin	A mobile-friendly skin that i
Alalike	Loosely based on the A.Lis
Amber	responsive mobile-friendly
Barthelme	A minimalist skin where wh
Blue Freedom	A White skin with a blue he
Blix	Blix is a two-column theme
Blog Skin	A blog-oriented skin that er
BlueBerry	xhtml 1.1 and css 2.0 confi
BlueFur	Blue Fur skin has kind of a
Bluehead	Elegant looking white skin
Blues	This skin has the blues ()
Boira	Boira skin (20111202 Alpha)
Bonny	Bonny is a fully CSS-driver
BS-001 PmWiki skin	This is a dark theme intend
Bs-002	Bs-002 Skin - This is a ther
Choice	Choice is a two-column the

3. In this documentation, I use NewsPaper skin so we can click the zip file on Download section

PMWIKI

THIS IS WORDPRESS NEWSPAPER THEME CONVERTED TO PMWIKI

1024

AGE

BOX

ENT

ES

OK

ATION

ENTS

18)

.

IG

S

S

ES

IONAL

T

i

JP

ONLY

ITTING

GATION)

A

STS

TING

ABLES

ING

NEWSPAPER

Neutral < Skins > Night (talk | users | test page | RWD? | not as default | unset | validate: HTML, CSS)

Summary: WordPress skin adapted for PmWiki.

Version: 2006-10-02

Prerequisites: PmWiki 2.1

Status: maintained

Maintainer: CarlosAB

Categories: Skins, PHP72

Download: [NewsPaper-2006-10-02.zip](#)^Δ

See Discussion at: [NewsPaper-Talk](#)

Contents [hide]

[Description](#)

[Download](#)

[Install](#)

[Notes](#)

[Release Notes](#)

Description

This is a WordPress Skin that was adapted for PmWiki .

If you are going to use in a UNIX environment, be sure to pay attention to permissions and ownership. The files inside the zip file have none of those set.

Download

Download: [NewsPaper-2006-10-02.zip](#)^Δ

PAGE A

VI

EE

HIS

ATT

PR

RECENT

SITE R

CHA

GROUP

CHA

GROUP

SK

NEWS

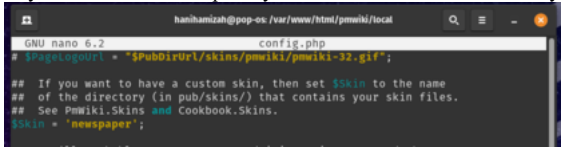
BACK

NEWSPAP

SKINSG

4. Download the skin into /var/www/html/pmwiki/pub/skins using command
sudo wget https://www.pmwiki.org/pmwiki/uploads/Skins/NewsPaper-2006-10-02.zip and unzip the file using sudo unzip NewsPaper-2006-10-02.zip

5. After we are sure there is newspaper skins directory in pub/skins, we can go to local/config.php again to configure PmWiki appearance here. Click CTRL + W \$skin = 'yourskinname' and replace your skin name with skin's name you downloaded just now.



6. We can change header text in skin by going to file newspaper.php and edit the php file.



```
Workspaces Applications Jan 19 5:27 PM
hanihamizah@pop-os: /var/www/html/pmwiki/pub/skins/newspaper
GNU nano 6.2 newspaper.php
<?PHP

global $WikiSlogan,$FmtPV;

$WikiSlogan = "Hani's Project for System Administration and Maintenance";
$FmtPV['$NpTime'] = 'strftime( "%b %d, %Y" , time())';
```

Retrieved from <http://192.168.56.105/pmwiki/index.php?n=Main.MainContent>
Page last modified on January 21, 2024, at 02:00 pm